Fail Closed

No approval, no signature.

An independent co-signer for crypto wallets. Unusual withdrawals run in an isolated sandbox before anything is signed.

Built on Vultr Serverless Inference + Cloud Compute

THE PROBLEM

Two real attacks. The signer signed what it was asked.

Binance · 29 Oct 2026 VERIFY

Hot wallet delegation drain (EIP-7702)

- About $6.3 billion VERIFY left hot wallets on about 10 chains VERIFY.
- One transaction carried 50 delegation authorizations VERIFY: the wallet was handed to the attacker's contract.
- No private key leaked VERIFY. A compromised internal service asked, and the signer signed.

Mercado Bitcoin · 2 Nov 2026 VERIFY

Multi-chain drain after key theft

- No delegation trick: plain signed transfers.
- One address was paid on 10 chains VERIFY, about 185 tokens VERIFY, one by one.
- Each transfer looked normal on its own.

Fail Closed is designed to reject this class of attack. We do not claim it would have stopped these incidents.

THE COST NOBODY COUNTS

After the attack, everyone waits.

- Both exchanges stopped withdrawals **for all customers** VERIFY, for days VERIFY.
- Honest customers pay for the attack too: their money is stuck.
- A full stop is the only tool when nothing narrower exists.
- **A guard should stop the attacker, not the business.**

Fail Closed is designed to reject this class of attack. We do not claim it would have stopped these incidents.

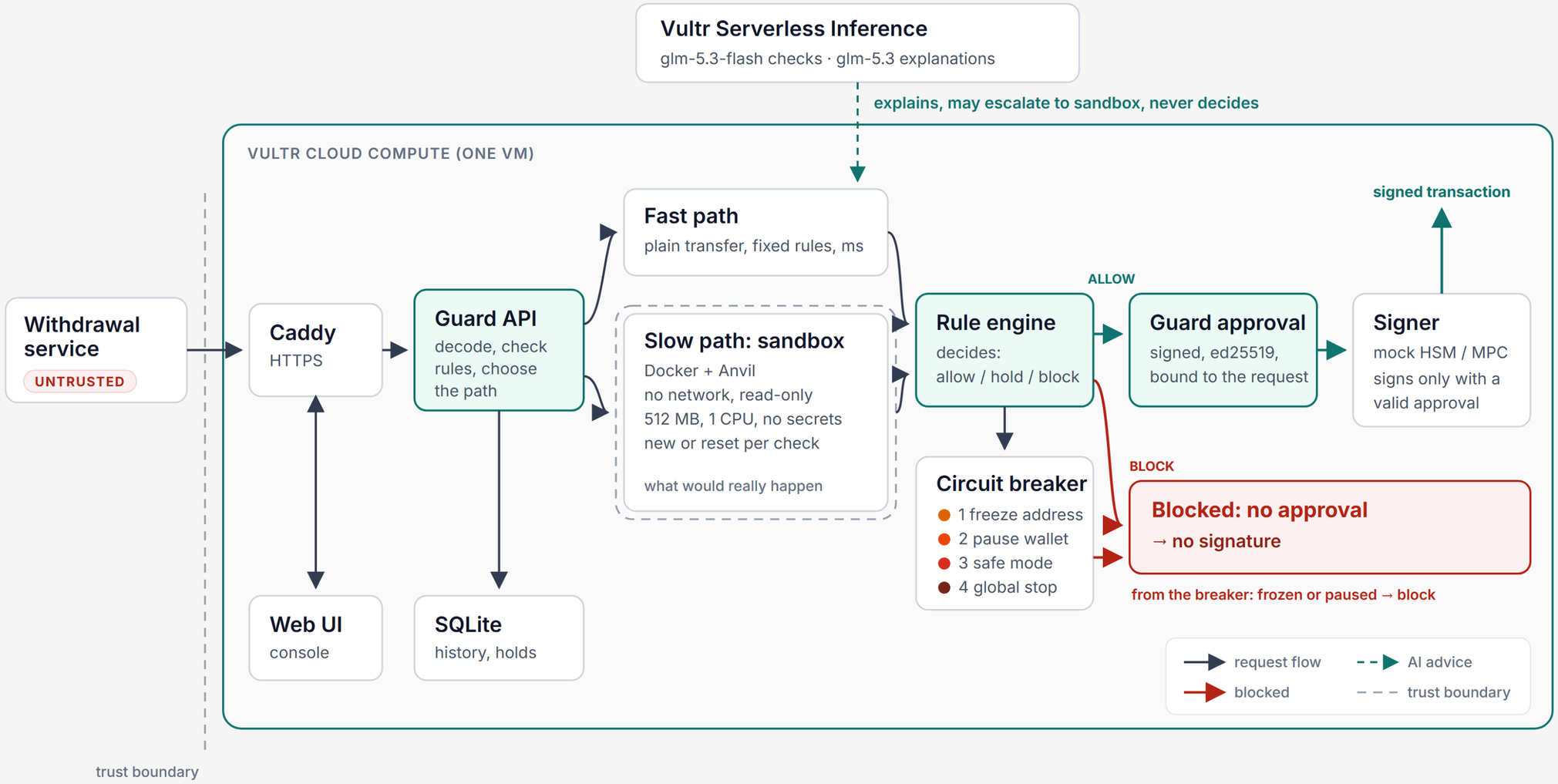
THE IDEA

Check what a signature will do, not who asks.

- **An independent co-signer.** The signer signs only with a Guard approval for that exact request.
- **Unusual requests run first** in an isolated sandbox: no network, no secrets, fake funds.
- **Fixed rules decide.** The AI explains in plain words. It may ask for a deeper check, never decides.
- **Fail closed.** Any error, timeout or empty result: no approval, no signature.

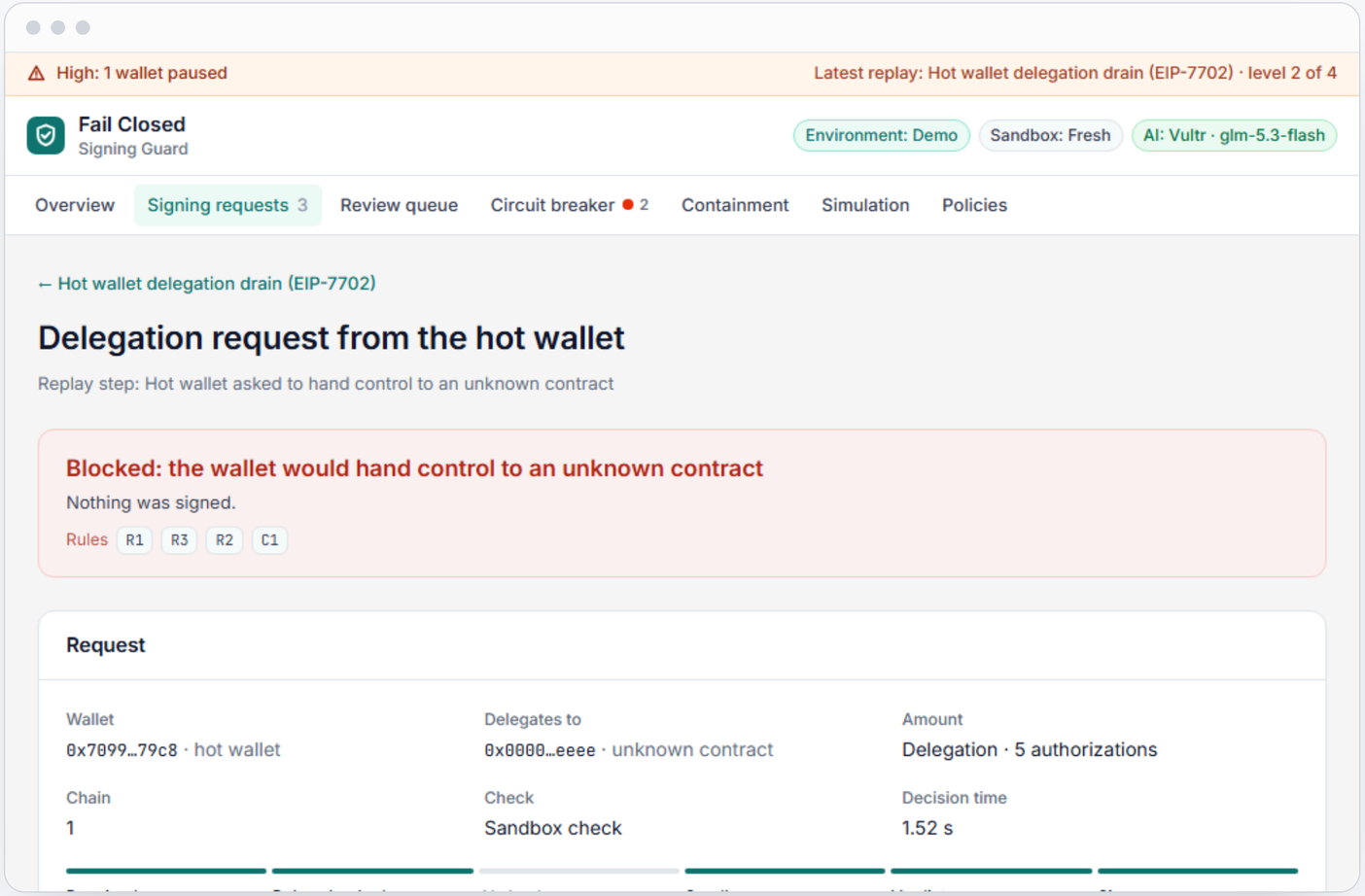
Fail Closed: how a signing request is checked

No approval, no signature. Fixed rules decide; the AI only explains.



DEMO 1 · HOT WALLET DELEGATION DRAIN

Blocked. Nothing was signed.



- The sandbox shows the wallet would be **handed to an unknown contract**.
- A test call as the attacker **moves 100 ETH** out (fake funds, in the sandbox).
- No approval, so **the signer refuses**. The wallet is paused.

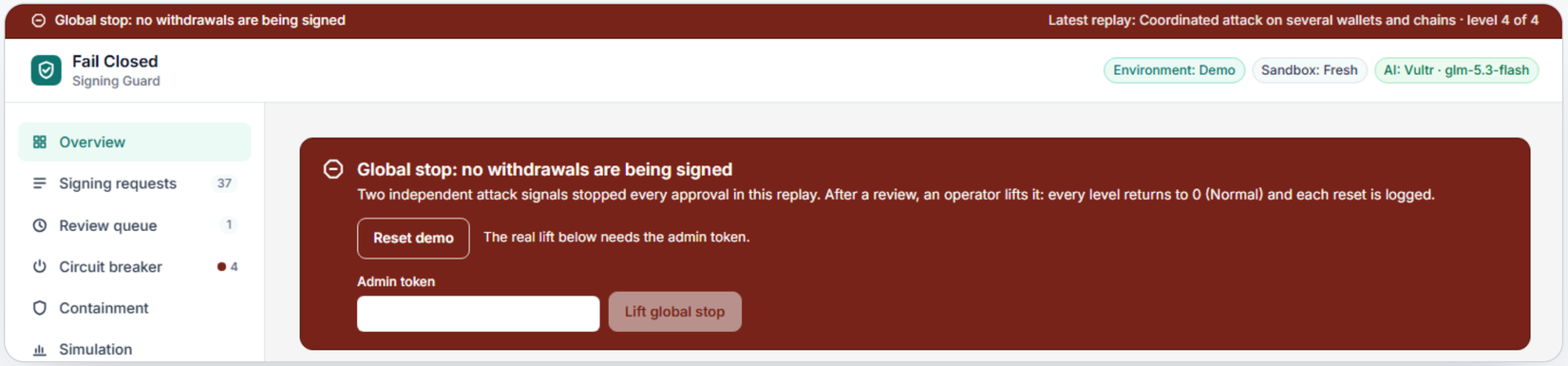
DEMO 2 · COORDINATED ATTACK

The risk level climbs 0 → 4, one narrow step at a time.

Before: level 0, Normal



After the attack: level 4, Global stop



- Level 1 freezes **one address**.
- Level 2 pauses **one wallet**.
- Level 3: **one chain** in safe mode.
- Other customers keep being paid until level 4.
- Level 4 needs **two independent signals**. Only an operator lifts it, with the admin token.

MEASURED RESULTS

500 normal withdrawals + 1 hidden attack

Attack stopped	**yes, in all 4 setups**
Normal customers refused	**0 of 500**
Fast path (plain payout to a known customer)	**3 ms median**
Slow path (full sandbox check)	**1.3 s median (warm) · 2.4 s (fresh)**
Same check with the real Vultr model (49 + 1)	**attack stopped · 0 refused · 0 rate limits**
Red team: hostile code inside the sandbox	**7 of 7 contained**

Measured on a laptop (Windows 11, 16 CPUs, Docker Desktop), 6 Nov 2026, with backend/cli.py loadtest: mock AI (800 ms per call) for the 500 + 1; real Vultr glm-5.3-flash for the 49 + 1. Red team: the browser test run of 6 Nov 2026. Not yet measured on the Vultr VM. Data: docs/results/.

BUSINESS VALUE

Protect the wallets without stopping normal customers.

- Everyday payouts to known customers are approved **in milliseconds**.
- New destinations and unusual requests get a **full simulation** first, in about a second or two.
- Large or odd payouts go to **a person**, not to a blanket stop.
- The breaker stops **as little as possible**: address, then wallet, then chain.
- In our load test: **0 of 500** normal customers refused (laptop, mock AI, 6 Nov 2026).

Honest limits

- A hackathon prototype: local test chain, fake funds.
- It does not stop the first break-in. It stops the break-in from getting a signature.
- If an attacker takes over the Guard or its key, the protection is gone: run it separately.
- Numbers so far come from a laptop, not yet from the VM.

What's next

- A real co-signer in MPC or multisig wallets, and in HSMs.
- Non-EVM chains.
- Simulation on a mainnet fork.
- A watchtower that spots a hijacked wallet and stops its top-ups.

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